

DATA 505 (Statistics Using R): Exam 1

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze data about Antarctic penguins from the Palmer Station research program. The data are available in the `palmerpenguins` R package.

```
# Install and load the required packages
if(!"palmerpenguins" %in% installed.packages()){
  install.packages("palmerpenguins")
}
if(!"ggplot2" %in% installed.packages()){
  install.packages("ggplot2")
}

library(palmerpenguins)
library(ggplot2)

# Load the penguins data
data("penguins")
```

Data Exploration (4 points)

1 (2 points)

Use the `str()` function to examine the structure of the penguins dataset.

```
str(penguins)
```

```
tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm  : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
 $ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

How many observations and variables are there?

Answer: The dataset penguins has 344 observations and 8 variables, including: species, island, bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, sex, year .

2 (2 points)

Extract the `bill_length_mm` column from the penguins dataset and store it in a new variable called `bill_lengths`. Then use the `summary()` function to examine this variable.

```
bill_lengths <- penguins$bill_length_mm
summary(bill_lengths)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
32.10	39.23	44.45	43.92	48.50	59.60	2

Research Question

We want to investigate whether the mean bill length of Adelie penguins differs significantly from 39 mm. This research question is motivated by previous studies suggesting that 39 mm might be a typical bill length for this species.

1 (4 points)

Filter the dataset to include only Adelie penguins and store the result in a new variable called `adelie_penguins`. Remove any rows with missing bill length measurements.

```
adelie_penguins <- penguins[penguins$species == "Adelie" & !is.na(penguins$bill_length_mm), ]
adelie_penguins
```

```
# A tibble: 151 x 8
  species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
  <fct>    <fct>          <dbl>         <dbl>          <int>        <int>
1 Adelie  Torgersen        39.1           18.7            181         3750
2 Adelie  Torgersen        39.5           17.4            186         3800
3 Adelie  Torgersen        40.3            18            195         3250
4 Adelie  Torgersen        36.7           19.3            193         3450
5 Adelie  Torgersen        39.3           20.6            190         3650
6 Adelie  Torgersen        38.9           17.8            181         3625
7 Adelie  Torgersen        39.2           19.6            195         4675
8 Adelie  Torgersen        34.1           18.1            193         3475
9 Adelie  Torgersen        42            20.2            190         4250
10 Adelie Torgersen        37.8           17.1            186         3300
# i 141 more rows
# i 2 more variables: sex <fct>, year <int>
```

2 (3 points)

Extract the bill lengths from the Adelie penguin data and store them in a variable called `adelie_bills`. How many Adelie penguins have complete bill length measurements?

```
adelie_bills <- adelie_penguins$bill_length_mm
length(adelie_bills)
```

```
[1] 151
```

Number of Adelie penguins with complete data: The number of Adelie penguins with complete data is 151 .

3 (5 points)

Calculate the following descriptive statistics for Adelie penguin bill lengths: - Mean - Standard deviation

- Median - 25th and 75th percentiles

```
mean(adelie_bills)
```

```
[1] 38.79139
```

```
sd(adelie_bills)
```

```
[1] 2.663405
```

```
median(adelie_bills)
```

```
[1] 38.8
```

```
IQR(adelie_bills)
```

```
[1] 4
```

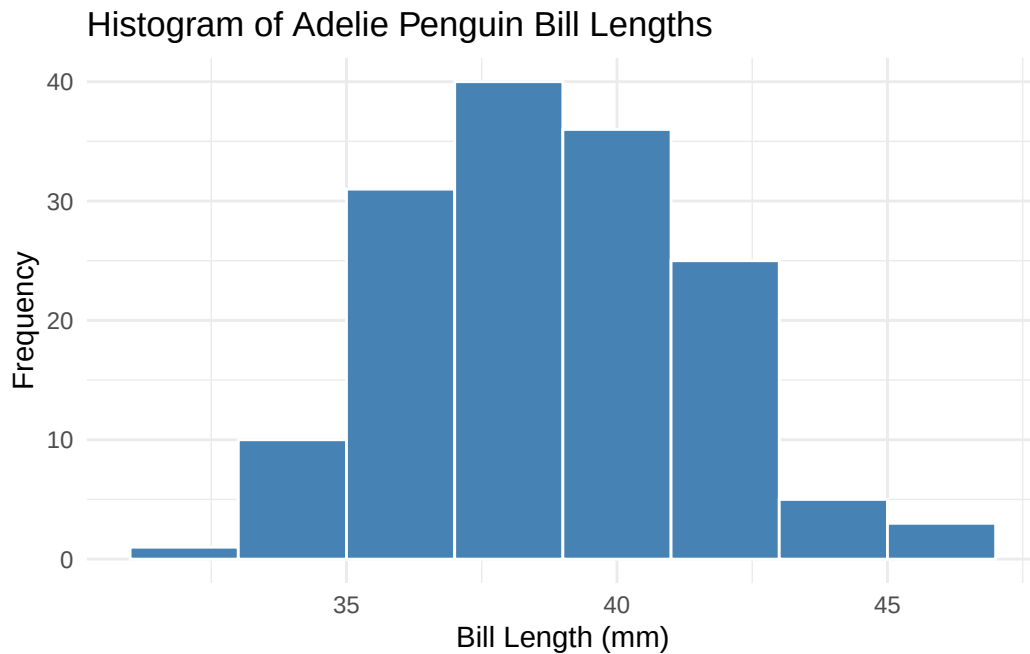
```
quantile(adelie_bills, probs = c(0.25, 0.75))
```

```
      25%      75%  
36.75 40.75
```

4 (5 points)

Create a histogram of Adelie penguin bill lengths using ggplot2 or base R. Include appropriate axis labels and a title.

```
ggplot(adelie_penguins, aes(x = bill_length_mm)) +
  geom_histogram(
    binwidth = 2,
    fill = "steelblue",
    color = "white") +
  labs(
    title = "Histogram of Adelie Penguin Bill Lengths",
    x = "Bill Length (mm)",
    y = "Frequency") +
  theme_minimal()
```



Based on the histogram, briefly describe the distribution of bill lengths (address shape, center, and spread):

Description: The distribution of the histogram of the Adelie penguin bill lengths is roughly symmetric and unimodal, with the center of the distribution is 38.8 mm. The middle 50% of Adelie Penguin have bill lengths between 36.75mm to 40.75mm, giving a sense of spread.

Statistical Analysis

1 (5 points)

Now we will formally test whether the population mean bill length μ differs from 39 inches, i.e. we will test:

$$H_0 : \mu = 39$$

against

$$H_A : \mu \neq 39$$

Perform a two-sided t-test to test these hypotheses. Use $\alpha = 0.05$.

```
test_result <- t.test(  
  x = adelic_bills, # numeric vector of observations  
  mu = 39, # value from H0  
  alternative = "two.sided"  
)  
test_result
```

One Sample t-test

```
data:  adelic_bills  
t = -0.96246, df = 150, p-value = 0.3374  
alternative hypothesis: true mean is not equal to 39  
95 percent confidence interval:  
 38.36312 39.21966  
sample estimates:  
mean of x  
 38.79139
```

Based on your test results, what is your conclusion? Interpret the results in the context of the research question.

Conclusion: The p-value is greater than 0.05. We fail to reject null hypothesis (H_0). There is no evidence in this sample that the population mean bill length differs from 39 inches. The Interpretation: We are 95% confident that the population mean bill length is between 38.36 and 39.21

2 (4 points)

Define another variable (a numeric vector) `confidence_interval` and assign as its value the two bounds (lower and upper) of a 95% confidence interval for the mean bill length of Adelie penguins. Print the result. Full points are possible only if you do not “hard code” the bounds for the interval, i.e. if you create this vector directly by using R functions applied to the dataset. Half points maximum if you define the vector by typing the entries manually.

```
test_result <- t.test(  
  x = adelia_bills,  
  mu = 39,  
  alternative = "two.sided",  
  conf.level = 0.95  
)  
  
confidence_interval <- test_result$conf.int  
confidence_interval
```

```
[1] 38.36312 39.21966  
attr(,"conf.level")  
[1] 0.95
```
